



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000
WEEK 4.04

TUTOR GUIDE

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The cases and materials in this guide were reviewed by;

- ☐ The curriculum management committees' critical care working group. Last reviewed Nov 2010.

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RELEVANT PRIOR LEARNING

MED 1000

1.28 Anatomy of the upper limb, brachial plexus (L & P)

MED2000

2.19 Pharmacology of local anaesthetic drugs (L)

2.23 Peripheral nerve blocks (L & P)

WEEK 04_ ANAESTHESIA		
Yr	Wk	LEARNING OBJECTIVES
4	4	- Define (conscious) procedural sedation. - Know the indications, contraindications and potential adverse effects for procedural sedation. - Know the key components of the history and examination when assessing a patient for procedural sedation. - In a table summarise the dosage, route of administration and duration of effect of the following medications used in procedural sedation, short acting benzodiazepines [eg midazolam], Opioids [eg fentanyl, morphine], Propofol, Ketamine and Nitrous Oxide. - Be able to prepare a patient for procedural sedation by knowing the equipment and facilities that are required. - Outline the management of patient following the administration of sedation for a procedure.
4	4	- Know the key components of the assessment of an elderly patient for an emergency anaesthetic. - Know the relevant anatomy for administering spinal anaesthesia. - Outline the process of spinal anaesthesia. - Detail the advantages and disadvantages of spinal anaesthesia. - Know the contraindications to spinal anaesthesia. - Outline the factors that predispose to post operative delirium and cognitive dysfunction.
4	4	- Outline the analgesic options available during labour. - Know the relevant anatomy for administering an epidural anaesthetic. - Describe the differences between a spinal and epidural anaesthetic. - Outline the advantages and disadvantages of epidural analgesia during labour. - Outline the potential risks and possible adverse effects of lumbar epidural analgesia in labour. - Know the indications for general anaesthesia for a LSCS.
4	4	- Review the anatomy of the brachial plexus and the major branches in the upper limb and be able to explain the relevant anatomy in performing regional upper limb anaesthesia. - Know the advantages and disadvantages of using regional upper limb anaesthesia. - Detail the prerequisites for safe regional anaesthesia.

CASE ONE

Short case number: 4_04_01

Category: Clinical Skills_Pharmacology

Discipline: Anaesthesia

Setting: Operating Theatres_rural

Topic: Regional Anaesthesia_Procedural Sedation

CASE

Mrs Smart is a 55 year old female who has presented to hospital today for a routine, elective colonoscopy. She has a past history of hypercholesterolemia. She is currently taking Simvastatin 20mg Nocte. She has no known medication allergies, and is otherwise well.

QUESTIONS

1. This procedure requires procedural sedation. What is procedural (conscious) sedation?
2. Outline the indications for procedural (conscious) sedation.
3. What are the important components of the history and examination when assessing this patient for procedural sedation?
4. What equipment and facilities are required to undertake safe procedural sedation?
5. Outline the **contraindications** and potential adverse effects of procedural sedation.
6. Outline the medications used in procedural sedation, including mechanism of action, dosage, route of administration and duration of effect.

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.
- Fulde GW. Emergency Medicine, the principles of practice. 5th Edition. Churchill Livingstone Elsevier. 2009

ANSWERS

This procedure requires procedural sedation. What is procedural (conscious) sedation?

Procedural sedation, also referred to as conscious sedation, is the administration of sedative drugs to facilitate performance of a distressing or painful procedure (Fulde, 2009, p.29).

“Procedural sedation and analgesia refers to the technique of administering sedative or dissociative agents, with or without analgesics, to induce a state that allows the patient to tolerate unpleasant or anxiety provoking procedures, whilst maintaining cardio-respiratory function” (Cameron et al, 2009, p.704).

Outline the indications for procedural (conscious) sedation.

Procedures involving significant pain (and/or anxiety). Dependent on the individual and tolerance levels, i.e. some adults are able to undergo shoulder dislocation reduction without sedation, and minimal analgesia; children more likely to need mild sedation for painful procedures such as suturing than adults.

Procedural sedation in anaesthesia:

- Colonoscopy
- Bronchoscopy

Examples of Procedures in Emergency:

- **Reduction of significant joint dislocations** – particularly when neurovascular compromise is present
- **Reduction and splinting of long bone fractures** – reduce blood loss
- **Painful / anxiety provoking procedures** – lumbar puncture, central line insertion, burns dressing, chest drain insertion, incision and drainage procedures
- **Cardioversion**

What are the important components of the history and examination when assessing this patient for procedural sedation?

Pre-Procedure Risk Assessment

1. Age
2. American Society of Anaesthesiologists (ASA) Classification – Class I and II are usually considered appropriate for Procedural Sedation

Table 22.3.1: ASA Classification

Class	
1	Healthy Patient, no medical problems
2	Mild Systemic Disease, i.e. Hypertension
3	Severe Systemic Disease, not incapacitating
4	Severe Systemic Disease, constant threat to life
5	Moribund expected to live <24hr irrespective of operation

3. Airway Assessment – Mouth opening, pharyngeal visualisation (Mallampatti score), neck movement, thyromental distance, dentition, history of difficult airway

Table 22.3.2: Airway assessment predictors of a difficult endotracheal intubation

1	Mallampatti III & IV
2	Inability to open mouth >4cm
3	Thyromental distance <6cm
4	Limitation of neck movement
5	Difficult in protruding lower jaw
6	History of difficult intubation

4. Appropriate medical, surgical, medication and allergy history

5. FASTING HISTORY – ensure CORRECT time of fasting, ask specifically about fluids and solids and exact time of last ingestion. ASA recommends at least 2 hours from last clear fluid intake, and at least 6 hours from last solid food intake.

6. Physical Examination – Cardiovascular, respiratory at a minimum

7. Informed Consent

What equipment and facilities are required to undertake safe procedural sedation?
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PROPER PLANNING PREVENTS POOR PERFORMANCE

AREA

Needs to be undertaken in an area with appropriate resuscitation equipment and monitoring

- Resuscitation bay in Emergency Department
- Operating Theatre
- Intensive Care Unit

EQUIPMENT

Need all equipment necessary for AIRWAY, BREATHING AND CIRCULATION management

- Airway and Breathing equipment: Including basic airway equipment (oxygen masks, tubing, supply, oropharyngeal airway, nasopharyngeal airway, bag valve mask, suctioning equipment), intubating equipment (laryngoscope, Endotracheal tubes, bougie, laryngeal mask airways, difficult airway equipment)

- Circulation Equipment: IV fluids, IV access devices (IVC, Intraosseus)

MONITORING

Need appropriate continuous cardiac and respiratory monitoring: ECG, respiratory rate, Oxygen Saturations, Carbon Dioxide monitoring, Blood pressure

Verbal and tactile stimulation of the patient to assess depth of sedation

Monitoring should continue post procedure until patients have returned to their baseline level of functioning.

STAFF

Need a medical officer with appropriate high level experience in sedation and airway management (ACEM recommends at least one advanced trainee or consultant), especially difficult airway management. Also appropriately trained and numbers of nursing staff to assist and monitor.

DRUGS

Need appropriate drugs available for sedation, as well as emergency drugs for intubation / circulation.

DOCUMENTATION

Patient consent

Documentation of timing of drugs given, vital signs and actual procedure undertaken

Outline the contraindications and potential adverse effects of procedural sedation.

Contraindications

Not fasted

- 6 hours for solids
- 2 hours clear fluids

Outline the medications used in procedural sedation, including mechanism of action, dosage, route of administration and duration of effect.

Drugs used in Procedural Sedation			
Drug	Dose	Duration	Side Effects
Midazolam	0.02 – 0.1mg/kg IV 0.05mg/kg PO	30 min IV 45-60 min PO	Loss of Airway Protection Respiratory Depression
Fentanyl	2-3 micrograms/kg IV	20-30 min	Respiratory depression Bradycardia
Propofol	0.5-1mg/kg IV	10 min	Pain on injection Seizures Decreases Blood Pressure Loss of airway protection
Ketamine	1-2mg/kg IV 3-5mg/kg IM	15 min IV 30 min IM	Emergence reaction, hallucinations

			Excessive Salivation Increased Intra-ocular and intracranial pressure
Nitrous Oxide	Given as 30-50% mix in Oxygen		Causes Gas expansion of gas filled structures (i.e. Pneumothorax)

From Fulde, G (2009), p.31

CASE TWO

Short case number: 4_04_02

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Emergency Department_Urban

Topic: Regional Anaesthesia_spinal

CASE

You are the Anaesthesia resident; the orthopaedic registrar has just called and asked you to book Mr Douglas Pixler for an emergency operation for his fractured neck of femur.



Mr Douglas Pixler is an 88 year old male from the local nursing home. He has sustained an undisplaced fracture of his right neck of femur [NOF] following a fall.

On reviewing the medical records you note an intercurrent medical history of

- Hypertension
- Chronic AF.

The admitting intern has also noted mild short term memory loss. He is an ex-smoker of 10 years – previous 30 pack year history.

QUESTIONS

1. Describe your approach to assessing and preparing this patient for surgery.
2. You present the patient to the consultant Anaesthetist who recommends the procedure be performed under a spinal anaesthetic. What is a spinal anaesthetic?
3. As you are preparing the patient the medical student asks you what the anatomical landmarks are for a spinal anaesthetic. What would you explain to the medical student?
4. You are explaining the procedure to Mr Pixler and his daughter, who asks you what the advantages are to performing the procedure under a spinal, rather than a general anaesthetic. What would you explain to her?
5. In addition to the advantages, outline the disadvantages and contraindications to spinal anaesthesia.
6. One of the concerns of the team looking after Mr Pixler is that he may develop post operative delirium. What is post-operative delirium? List likely causes in this scenario?

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.

Advanced Reading:

- Halaszynski T. Pain Management in the Elderly and Cognitively Impaired Patient: The Role of Regional Anaesthesia. *Current Opinion in Anaesthesiology* 2009;22:594-599
- Silverstein J et al. Central Nervous System Dysfunction after Noncardiac Surgery and Anesthesia in the Elderly. *Anesthesiology* 2007;106:622-8

ANSWERS

Describe your approach to assessing and preparing this patient for surgery

The anaesthetist should attend the patient to perform the usual pre-anaesthetic consultation with particular emphasis on:

- (i) History of the presenting complaint: it is critical to determine whether the fall causing the fracture occurred as a consequence of a cardiac or neurological event such as angina, syncope or stroke
- (ii) Premorbid level of function: this pertains to both physical and cognitive function as hospital admission in the elderly population often reveals patients struggling and/or failing to cope in their current living situation. Any reduction in their ability to function following surgery will also impact on the level of care they will require on discharge.
- (iii) Past Medical and Surgical History: particular emphasis on cardioresp illness, GORD, and systems review.

As usual, a comprehensive assessment is completed with history of previous anaesthetic problems, assessment of exercise tolerance and fasting status, history of smoking, alcohol intake, current medication and allergies.

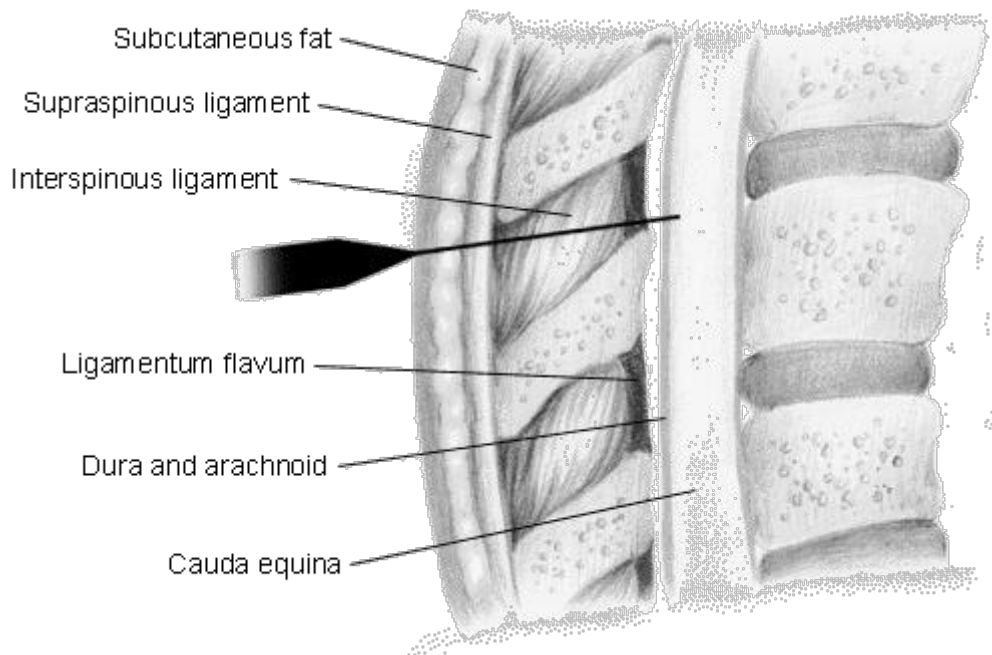
Physical examination should assess indicators of difficulty with airway management, condition of teeth, observations, hydration, venous access, cardiorespiratory examination and other system examination as indicated.

Routine investigations should include FBC, U and E's, CXR and ECG.

Patients will be bed-bound in traction until operative fixation of fractured neck of femur (#NOF). Prompt surgery and early mobilization is associated with improved outcomes in elderly patients as this avoids long periods of supine immobility that can promote lung atelectasis, poor respiratory secretion clearance, silent aspiration, pneumonia, pressure sores and deep venous thrombosis (DVT). The role of the anaesthetist is to optimise the patient's physiology (eg. hydration) and the management of any pre-existing conditions (eg. heart rate control for chronic AF).

What is a spinal anaesthetic?

Using a single injection technique, local anaesthetic (LA) is introduced into the subarachnoid space and distributes in the CSF to cause a temporary conduction block of the spinal nerves. Opioids and other agents can be used in conjunction with LA to enhance the block.



The spinal injection is performed using aseptic technique in the lumbar spine, as low as is practical and always below L1 level to avoid direct trauma to the spinal cord.

Standard combinations of bupivacaine and fentanyl onset in approx 5-10mins and provide an anaesthetic block for the lower body for approx 90mins.

Patients remain spontaneously breathing and this technique can be used with or without additional sedation.

You are explaining the procedure to Mr Pixler and his daughter, who asks you what the advantages are to performing the procedure under a spinal, rather than a general anaesthetic. What would you explain to her?

Advantages

- (i) avoids the need to manage the airway and provide PPV which lowers the risk of respiratory complications such as atelectasis, aspiration pneumonia
- (ii) provides good early post-op analgesia
- (iii) reduces the amount of drugs to which the patient is exposed, hence may have a role in reducing the risk of post-operative cognitive dysfunction
- (iv) reduces surgical stress response
- (v) reduces risk of lower extremity DVT

In addition to the advantages, outline the disadvantages and contraindications to spinal anaesthesia.

Disadvantages and Adverse Effects

- (i) hypotension secondary to lower body sympathetic block +/- bradycardia and decreased cardiac output
- (ii) post-dural puncture headache
- (iii) nausea and vomiting

- (iv) infection: abscess or meningitis
- (v) neurological injury: direct trauma, epidural haematoma or abscess, neurotoxic agents
- (iv) inappropriately high block or total spinal

Contraindications

Absolute

local infection at procedure site
sepsis/bacteraemia
coagulopathy
intracranial hypertension
fixed cardiac output state ie. severe aortic stenosis
patient refusal

Relative

pre-existing neurological deficits
previous lumbar spinal surgery
unco-operative patient

One of the concerns of the team looking after Mr Pixler is that he may develop post operative delirium. What is post-operative delirium? List likely causes in this scenario?

Elderly patients are at high risk of altered mental state in the perioperative period which is associated with increased morbidity and mortality. Delirium refers to changes in the first 24-72 hours post-op, whereas POCD persists for weeks to months after surgery. Incidence of delirium is reported as 5-15% in the elderly after general anaesthesia. Average incidence following surgery for hip fracture is 35%, but has been reported as high 62%

Predisposing factors include advancing age (70+years), sensory impairment, renal impairment, history of alcohol abuse, pre-operative opioids, low educational levels and prior cognitive deficit or cerebrovascular accident.

Other factors that influence its development include:

- pre-existing cognitive deficit
- surgery
- pain
- sepsis
- hypoxia, hypotension and metabolic disturbances
- polypharmacy and withdrawal states
- environment

These issues should be considered when trying to diagnose and manage mental state alterations in elderly post-op patients. Management should also be directed at preventing the patient from self-harm (eg. falls while wandering, trying to climb out of bed) with the use of appropriate nursing supervision, and physical and/or pharmacological restraints.

CASE THREE

Short case number: 4_04_03

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Labour floor_urban

Topic: Regional Anaesthesia_Epidural.

CASE



Sarah is a 33y.o. primiparous woman who is requesting insertion of a lumbar epidural for analgesia in labour. She is 40 weeks pregnant and has been in established labour for 5 hours. During this time she has used heat packs, positioning and Entonox gas for analgesia with modest effect.

Her last vaginal examination shows cervical dilation at 3 cm with occipito-posterior fetal presentation.

Her obstetrician is preparing to commence a syntocinon infusion to augment her contractions.

QUESTIONS

1. What are the physiological changes associated with late pregnancy?
2. Sarah is concerned generally about medications while she is pregnant. Outline the classification of drugs according to their safety in pregnancy.
3. Sarah has already tried several modes of analgesia, explain the range of analgesic options available for labour?
4. What is the difference between an epidural and spinal block? In particular outline the different regional anatomy.
5. What are the benefits of epidural analgesia in labour?
6. Explain the potential risks and adverse effects of lumbar epidural analgesia in labour?

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.

FURTHER CASE INFORMATION

Sarah has an Epidural Block procedure. After 14 hours of labour Sarah is still only 6cm dilated and there are decelerations on the Fetal Heart Rate monitor. The Obstetric team advise an urgent LSCS is indicated.

FURTHER QUESTIONS

7. What are the particular risks of general anaesthesia for delivery?
8. The epidural put in earlier has not been very effective. What options for anaesthesia are there for this patient?

ANSWERS

What are the physiological changes associated with late pregnancy?

CVS

- decrease SVR
- increase CO (SV and HR)
- increase blood volume (40% at term)
- right shift of oxyHb dissociation curve (improve O2 delivery)
- aortocaval compression in supine position (from approx 28/40)

Resp

- increase minute ventilation (approx 50% at term)
- increase O2 consumption
- decrease FRC
- decrease PaCO₂
- vulnerable to (1)airway closure during tidal ventilation and (2) hypoxia
- capillary engorgement in mucosa of resp tract

CNS

- decreased MAC
- decreased LA requirement for neuraxial anaesthesia

GIT

- delayed gastric emptying
- increased intragastric pressure
- decrease LOS pressure
- equals risk of regurgitation and aspiration
- reduced plasma cholinesterase activity

Haem

- hypercoagulable state

Renal

- increase GFR (normalizes towards term)
- increase creatinine clearance

SYSTEM	CHANGE IN PREGNANCY	COMMENTS
Metabolic System		
Basal body temperature	Increase	
Oxygen consumption	Increase	20%–30% Increase
Cardiovascular System		
Cardiac output	Increase	30%–50% Increase; peak effect at 20–24 weeks of gestation
Stroke volume	Increase	
Heart rate	Increase	
Peripheral resistance	Decrease	
Blood volume	Increase	40%–50% Increase; mostly plasma
Plasma volume	Increase	45%–55% Increase; dilutional anemia common
Red cell mass	Increase	20%–30% Increase
Urinary System		
Renal pelvic diameter	Increase	Hydronephrosis common
Ureteral diameter	Increase	Hydroureter common
Glomerular filtration rate	Increase	30%–50% Increase; peak second trimester
Renal blood flow	Increase	40% Increase
Respiratory System		
Respiratory rate	No change	
Total lung capacity	No change	Volumes redistributed second half of pregnancy
Tidal volume	Increase	By 200 mL
Residual volume	Decrease	By 300 mL because of enlarging uterus
Gastrointestinal System		
Lower esophageal sphincter tone	Decrease	Progressive decrease to term
Gastric emptying time	Increase	
Intestinal transit time	Increase	
Colonic transit time	Increase	
Gastric acid secretion	Increase	
Hepatobiliary System		
Gallbladder emptying time	Increase	
Liver size	No change	
Liver histology	Subtle change	Subtle, nonspecific, including cytoplasmic fat vacuoles

<http://www.msdlatinamerica.com/ebooks/TextbookofGastroenterology/files/e258b84b57041a772d055c14591700a2.gif>

Conditions	Change during pregnancy	Normal pregnancy values
Cardiovascular		
Heart rate	Increases 15–20 bpm	75–95 bpm
Cardiac output	Increases 30–50%	6–8 l/min
Mean arterial blood pressure	Decreases 10 mmHg in midtrimester	80 mmHg
Systemic vascular resistance	Decreases 10–15%	1200–1500 dynes/s/cm ⁵
Respiratory		
Tidal volume	Increased 40%	700 ml
Minute ventilation	Increased 40%	10.5 l/min
Expiratory reserve volume	Decreased 15–20%	550 ml
Functional residual capacity	Decreased 20–25%	1350 ml
Blood gas		
pH	Unchanged	7.4–7.45
pCO ₂	Decreased	27–32 mmHg
pO ₂	Increased	100–108 mmHg
HCO ₃	Decreased	18–21 mEq/l
Hematologic		
Blood volume	Increases 30–50%	4500 ml
Erythrocyte volume	Increases 10–15%	
Hematocrit	Decreased	32–34%
White blood cell count	Increased	5000–15,000/mm ³
Factors I, II, V, VII, VIII, IX, X and XII	Increased	
Fibrinogen	Increased	>400 mg/dl
Genitourinary		
Renal blood flow	Increased 50–60%	700 ml/min
Glomerular filtration rate	Increased 60%	140 ml/min
Serum creatinine	Decreased	<0.8 mg/dl
Serum urea nitrogen	Decreased	<13 mg/dl
<i>Bpm: Beats per min.</i>		
 Source: Women's Health © 2009 Future Medicine Ltd		

<http://img.medscape.com/article/701/797/701797-tab1.jpg>

Sarah is concerned generally about medications while she is pregnant. Outline the classification of drugs according to their safety in pregnancy.

Australian pregnancy categories (FROM Australian Medicines Handbook)

Category A

Drugs that have been taken by a large number of pregnant women and women of child-bearing age, without any proven increase in the frequency of malformations or other direct or indirect harmful effects on the fetus having been observed.

Category B1

Drugs that have been taken by only a limited number of pregnant women and women of child-bearing age, without an increase in the frequency of malformation or other direct or indirect

harmful effects on the human fetus having been observed. Studies in animals have not shown evidence of an increased occurrence of fetal damage.

Category B2

Drugs that have been taken by only a limited number of pregnant women and women of child-bearing age, without an increase in the frequency of malformation or other direct or indirect harmful effects on the human fetus having been observed. Studies in animals are inadequate or may be lacking, but available data show no evidence of an increased occurrence of fetal damage.

Category B3

Drugs that have been taken by only a limited number of pregnant women and women of child-bearing age, without an increase in the frequency of malformation or other direct or indirect harmful effects on the human fetus having been observed. Studies in animals have shown evidence of an increased occurrence of fetal damage, the significance of which is considered uncertain in humans.

Category C

Drugs that, owing to their pharmacological effects, have caused or may be suspected of causing, harmful effects on the human fetus or neonate without causing malformations. These effects may be reversible. Specialised texts should be consulted for further details.

Category D

Drugs that have caused, are suspected to have caused or may be expected to cause, an increased incidence of human fetal malformations or irreversible damage. These drugs may also have adverse pharmacological effects. Specialised texts should be consulted for further details.

Category X

Drugs that have such a high risk of causing permanent damage to the fetus that they should not be used in pregnancy or when there is a possibility of pregnancy.

Additional information about the Australian categories

For drugs in the B1, B2 and B3 categories, human data are lacking or inadequate and subcategorisation is therefore based on available animal data. The allocation of a B category does not imply greater safety than the C category. Drugs in category D are not absolutely contraindicated in pregnancy (eg anticonvulsants). Moreover, in some cases the D category has been assigned on the basis of suspicion.

Sarah has already tried several modes of analgesia, explain the range of analgesic options available for labour?

Principles are safety for mother and foetus, particularly avoidance of:

- respiratory depressant effects
- impairment of uterine contractility
- other adverse effects on the fetus, mother or progress of labour (ie NSAIDS)

Pharmacological

Parenteral opiates

- Intermittent opioid injection or PCA (remifentanyl)
- Pethidine most commonly used – nausea, foetal depression if given too close to delivery (peak foetal effect 2-3 hours after imi dose to mother)

Nitrous Oxide (50%) and Oxygen gas combination for inhalation

- Long history
- Nausea and disorientation
- Entonox efficacy user dependent Onset 45-60 secs so must start to use it as soon as contraction starts, not wait for it to become intense.

*both above complicated by nausea and vomiting and dysphoria

Regional analgesia

- neuraxial: epidural, spinal, CSE
- paracervical block

Non-Pharmacological

Heat

TENS

Distraction/Meditation techniques

Physical positioning

Acupuncture/Hypnotherapy

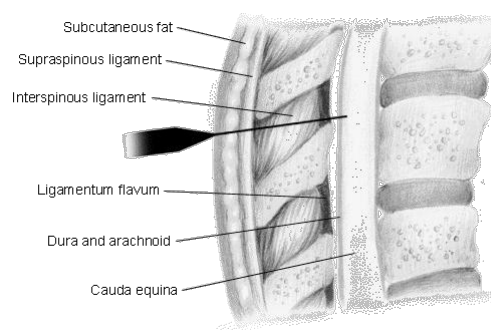
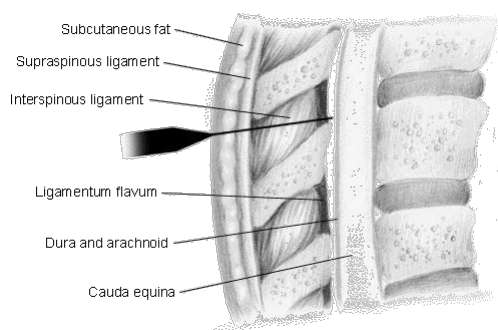
What is the difference between an epidural and spinal block? In particular outline the different regional anatomy.

A spinal block involves a single injection to instil LA +/- opioid into the **intrathecal space**. In contrast, an **epidural block identifies the epidural space outside the thecal sac**. Into this potential space **LA +/- opioid is instilled where it can access and cause conduction block** in the spinal nerves as they pass out the exit foraminae. The drugs can be instilled by single injection or infused via a catheter placed in the epidural space.

While it is preferable to instrument the epidural space below the conus medullaris (L1-2 level), thoracic and cervical epidurals can also be used for regional analgesia.

Cord ends L1 in adults, L3 in children. Need analgesia to T10 for uterine contractions, sacral segments for delivery. T4 for LSCS. Thecal sac ends S2 so from L1 to S2 can enter CSF space without risk of hitting cord. Surface anatomy – ASIS approx L3-4, but some variation. Spinal nerve roots bathed in CSF. Three layers of meninges – Pia mater (attached to neural structures; Arachnoid (spider-thin) and closely associated with dura, though there is a potential sub-dural space ; dura mater; outside dura is epidural space, containing fat and blood vessels; then ligamentum flavum, interspinous ligament. Nerve roots travel into epidural space where they can be exposed to LA, opiates or (occasionally) other drugs. The nerves are arranged in bundles in such a manner that, paradoxically, the larger sensory nerves are blocked more easily than the slightly smaller motor nerves. Insertion of epidural in a thin young person is usually not difficult, provided proper positioning, in foetal position. Can be done with patient sitting or lying in L Lateral position

(operator preference, though obese certainly easier sitting). Epidural space identified by loss of resistance, either to air or saline in a special low-resistance syringe attached to a Touhy needle. Catheter can be advanced up Touhy for repeat doses or infusion, or can be a “single shot”



What are the benefits of epidural analgesia in labour?

- (i) Maternal analgesia attenuates the increase in circulating catecholamines and stress hormones as well as the cyclical lowering of maternal pH associated with painful contractions
- (ii) Allows mother to rest and regroup for second stage labour
- (iii) Allows more controlled second stage labour which may assist in prevention of tears
- (iii) Recent evidence suggests a benefit for maintaining the integrity and function of pelvic floor muscles post-delivery
- (iv) Can be “topped up” to provide a block of sufficient height and intensity for emergency LSCS

Explain the potential risks and adverse effects of lumbar epidural analgesia in labour?

- (i) Post-dural Puncture Headache
- (ii) Nausea/Vomiting
- (iii) Pruritis
- (iv) Hypotension
- (v) Inadvertent intrathecal or intravascular injection of LA
- (vi) Inadvertent injection of incorrect substance into epidural space
- (vii) Failure of block
- (viii) Infection: abscess, meningitis
- (ix) Neurological event/injury: direct trauma from instrumentation or epidural haematoma formation

*abscess or haematoma may have delayed presentation post-delivery
any symptoms suggestive of sepsis or neurological deficit +/- back pain should trigger investigation for these rare but potentially devastating complications

The patient will require a urinary catheter and will not mobilize during the remainder of the labour once block is established. (While “walking epidurals” are possible, the intense nursing supervision required to eliminate risk of falls is generally not compatible with standard staffing ratios in labour wards.)

The effect of epidural analgesia on the progress of the labour is controversial. Studies have identified an association between epidurals and increased incidence of instrumented delivery. However, it is argued that greater pain and prolonged duration of labour accompany malpresentations that lead to the request for epidural analgesia in many of these cases.

What are the particular risks of general anaesthesia for delivery?

Due to physiological changes of pregnancy AND to any co-morbidity eg pre-eclampsia. At term Oxygen consumption increased 40%; FRC reduced so desaturate very quickly. Gastric emptying impaired – increased aspiration risk; supine hypotensive syndrome (aorto-caval compression – keep wedge under R hip). Airway oedema common also large breasts can combine to make intubation difficult. Always assume stomach not empty so need rapid-sequence induction with cricoid pressure to minimise aspiration risk. GA technique needs to keep in mind risk of neonatal respiratory depression so minimise opiates and BZD until cord clamped. Neuromuscular blockers highly polar so little placental transfer.

The epidural put in earlier has not been very effective. What options for anaesthesia are there for this patient?

It is urgent so probably best not attempt to reinsert EDB as it will be at least 20 mins even using 2% lignocaine. Alternatives are **GA** (see above for risks; other disadvantages are partner cannot be present at delivery); **Spinal anaesthesia** best option – effective in <5mins, single shot duration 2-3 hours, ideal for LSCS; Greater hypotension risk than epidural (see phenylephrine section above); partner can be present; low risk of PDPH with 26G needle; small amount intrathecal fentanyl gives several hours post-op analgesia; 0.5% bupivacaine 2.6mls or hyperbaric 2.2 mls usually reliably blocks to T-4 which is level ideal for LSCS

CASE FOUR

Short case number: 4_04_04

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Pre-admission clinic_urban

Topic: Regional anaesthesia_upper limb blocks

CASE



Bob is booked for repair of his severe and disabling left Dupuytren's contracture. He has been told by his hand surgeon that he will have "a block" from the anaesthetist.

You are interviewing Bob in the clinic for his preadmission workup and he asks you to explain what this means.

QUESTIONS

1. In explaining the anaesthetic to Bob, review the anatomy of the brachial plexus and its major branches in the upper limb.
2. Describe the sensory innervation of the upper limb.
3. What are the various approaches to brachial plexus regional anaesthesia?
4. What are the benefits and disadvantages of regional anaesthesia for hand surgery like that Bob is anticipating?
5. In preparing for this procedure, what are the prerequisites for safe regional anaesthesia?
6. Bob explains that he is anxious about being awake during the procedure, for all his previous operations he has been 'knocked out'. What therapeutic options could the anaesthetist offer to reduce his anxiety?

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.

Additional Resources

- Cousins and Bridenbaugh's Neural Blockade In Clinical Anaesthesia and Pain Medicine 4th Ed 2009

FURTHER CASE INFORMATION

You see Bob at 4 weeks post-op in clinic. He complains of persistent pain and swelling of the right hand with intermittent colour change and sweating. He is unable to make a fist or extend his fingers, and is avoiding using the hand.

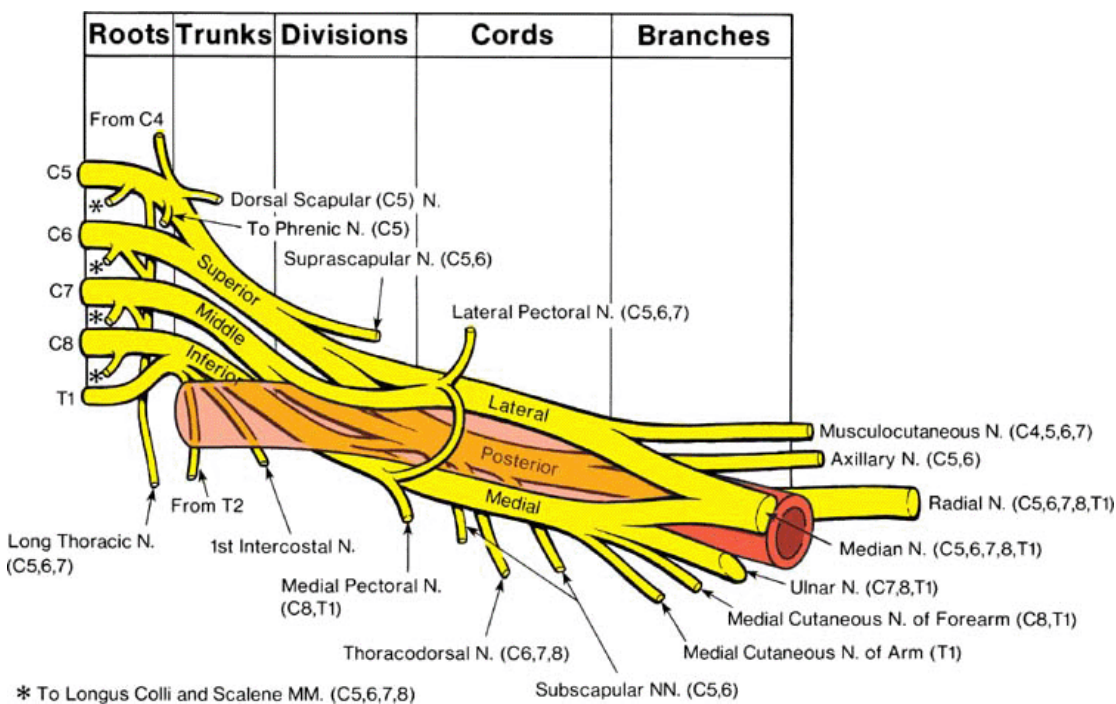
QUESTIONS

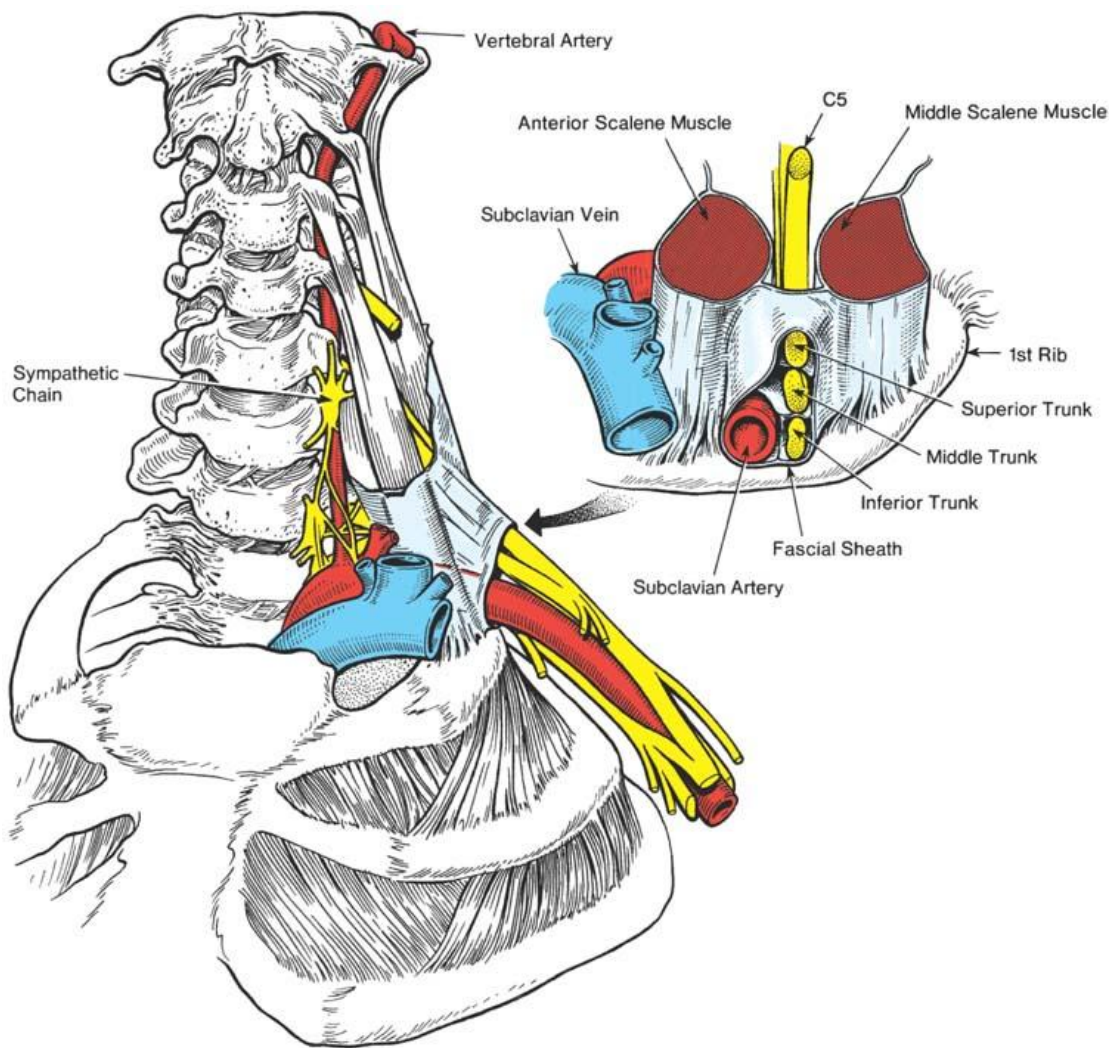
7. What do you think might be happening?

ANSWERS

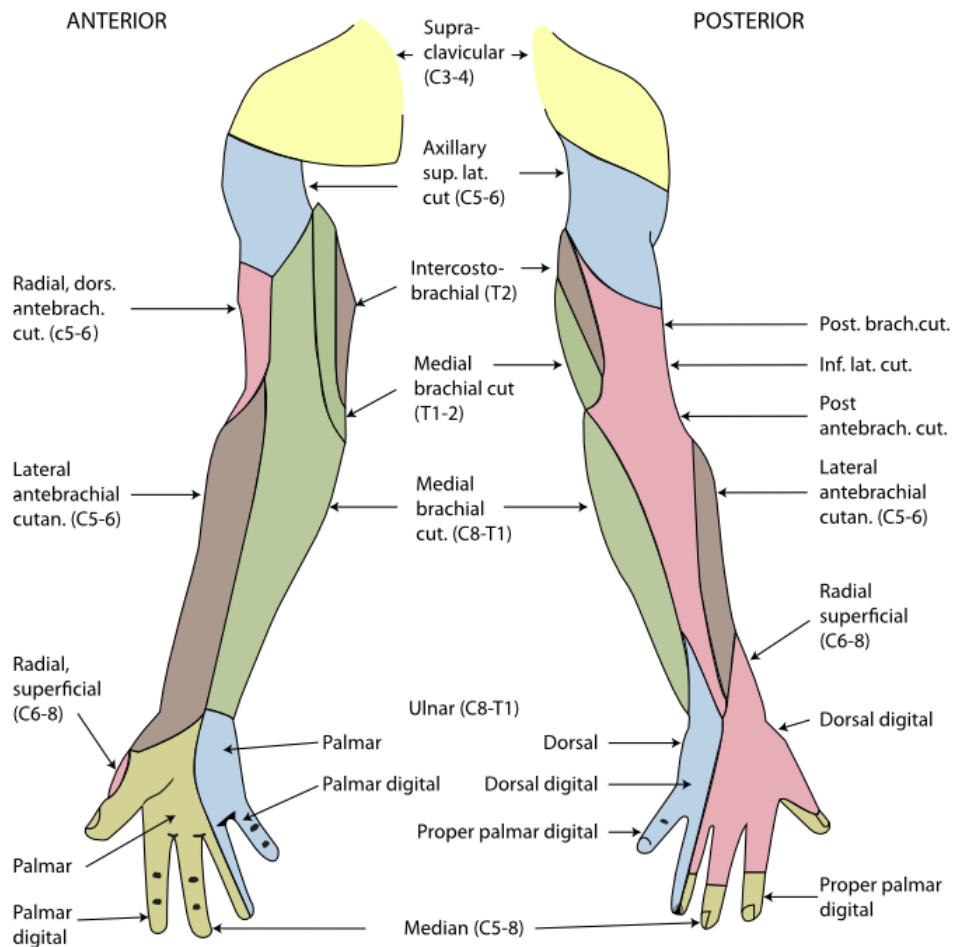
Review the anatomy of the brachial plexus and its major branches in the upper limb.

- formed from the ventral primary rami of C5 to T1, possible contribution from C4 and T2
- vertebral artery lies anterior to the nerve roots as they exit the intervertebral foramina
- roots invaginate the prevertebral fascia which forms an investing connective tissue sheath
- the roots traverse between the anterior and middle scalene muscles and converge into 3 vertically arranged trunks (superior, middle and inferior).
- at the level of the first rib, the trunks lie posterior, lateral and cephalad to the subclavian artery
- each trunk divides into ant and posterior divisions as they cross the lateral border of the first rib and merge into 3 cords (medial, lateral and posterior) that wrap around the second part of the axillary artery behind pectoralis minor
- more distally in the axilla, the terminal 4 nerves form and continue their close relationship to the axillary artery





Describe the sensory innervation of the upper limb.



<http://en.wikipedia.org/wiki/File:Gray812and814.PNG>

3. Approaches

- Interscalene
- Supraclavicular
- Infraclavicular
- Axillary

4. Benefits

- avoidance of GA and attendant risks
- excellent intra- and post-operative analgesia therefore opioid sparing
- sympathetic blockade may be beneficial for vascular surgery
- potential reduction in development of chronic pain

Negative aspects

- potential risks and complications eg. neural or vascular injury, LA toxicity
- incomplete or failed block necessitating conversion to GA
- patient intolerance of technique
- intolerance of occlusive tourniquet

5. Standard requirements for safe practice of regional anaesthesia

- informed patient consent
- appropriate location
 - ie. sterility, access to staff, emergency drugs and resuscitation equipment
- intravenous access
- devices to confirm appropriate position of needle
 - ie. ultrasound machine, nerve stimulator
- variety of local anaesthetics (ideally, choose best side effect profile for desired effect, administer in small repeated aliquots following aspiration to minimize inadvertent intravascular or other incorrect site administration)
- conscious and co-operative patient to assist with recognition of:
 - (i)inadvertent intravascular or other incorrect site administration
 - (ii)nerve injury
 (this can be achieved by use of light sedation only)

6. Light sedation is commonly used in conjunction with regional anaesthesia to enhance patient comfort and satisfaction. Low dose midazolam usually provides anxiolysis and amnesia sufficient for this purpose. The patient should be reassured of this and the capacity to deepen sedation if necessary, however inducing levels of sedation that compromise the airway or ventilation may negate a significant component of the benefit of opting for a regional technique.

7. Complex Regional Pain Syndrome (CRPS)

Bob's symptoms and signs may represent the onset of CRPS.

CRPS is a syndrome of pain, allodynia and hyperalgesia in body part in association with:

- an initiating noxious event/stimulus
- evidence at some time of oedema, altered skin blood flow (temp or colour change) and abnormal sudomotor activity (excessive or reduced sweating)
- skin, hair and nail quality changes
- motor changes (tremor, dystonia)

It is a diagnosis of exclusion. Key conditions to exclude are:

- infection
- fracture
- ischaemia
- inflammatory conditions (eg arthritides)
- erythromyalgia
- vasculitis/DVT
- painful peripheral neuropathy
- disuse atrophy and fear avoidance

Pathophysiology is not clearly understood. Evidence for contribution of neurological, inflammatory and immune-mediated mechanisms in genesis of CRPS. Further research required.

Risk factors include:

- female
- type of injury/surgery – high incidence with forearm fracture and TKR
- previous CRPS
- premorbid anxiety

Early recognition and intervention are crucial to optimizing outcome (hence important for junior medical staff to have basic knowledge of the syndrome). Ideally managed with referral to multidisciplinary pain management centre.

Optimising analgesia, functional restoration and addressing mood disorder/abnormal cognitions are the pillars of management. Some advanced regional techniques or implantable spinal cord stimulators may also be considered.

Patients with a history of CRPS should have this flagged to surgeon, anaesthetist +/- pain medicine specialist prior to further surgery on the affected or other body part as this will influence perioperative management.